

TECHNICAL SPECIFICATION: DISTRIBUTED CACHING LAYER v4.2

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1. Purpose & Architectural Overview

This document specifies the multi-tier distributed caching architecture designed to eliminate redundant LLM embedding computations and reduce database read queries on repetitive semantic queries. Target latency is sub-5ms for cache hits.

The cache employs a two-layer hierarchy: Layer 1 is in-memory local LRU cache (Python cachetools) with 5,000 item capacity. Layer 2 is a distributed Redis 7.2 cluster operating with cluster mode enabled and Raft-based consensus failover.

2. Cache Key Formulation & Eviction Policies

Cache keys are computed using SHA-256 over normalized text payloads and tenant UUIDs. The schema follows: `documind:tenant:{user_id}:hash:{sha256}`. TTL for document embeddings is strictly 14 days, whereas semantic query cache entries expire after 4 hours.

3. Circuit Breaker Parameters & Resilience

To shield upstream vector databases and relational databases during thundering herds, a Resilience4j-inspired sliding-window circuit breaker is enforced. The breaker transitions to OPEN if failure rates exceed 25% over a 60-second window with a minimum call volume of 40 requests.

When in OPEN state, the application falls back immediately to degraded degraded local lexical search without raising HTTP 500 exceptions to end users.

4. Performance Benchmark Targets

- Cache Hit Ratio target: $\geq 68\%$ across all production RAG workloads.
- Read latency: $P50 < 1.2\text{ms}$, $P95 < 3.8\text{ms}$, $P99 < 8.5\text{ms}$.
- Maximum allowable cluster memory utilization: 75% of available 64GB RAM.

5. Observability & Prometheus Metrics

All cache interactions expose Prometheus metrics on port 9090. Metric names:

- `documind_cache_hits_total{tier='redis'}`
- `documind_cache_misses_total{tier='redis'}`
- `documind_cache_latency_seconds_bucket`

Alert threshold: Page on-call if cache hit ratio drops below 40% for more than 15 consecutive minutes.

6. Disaster Recovery & Backup Plan

Redis RDB snapshots are taken every 3 hours and pushed to encrypted AWS S3 buckets in region us-east-1.

Append-Only File (AOF) with `fsync everysec` is enabled to limit potential data loss to under 1 second of cache keys.

7. Security & Encryption Standards

All Redis connections require mTLS with client certificate verification. Passwords must have at least 64 bytes of cryptographically random entropy loaded from HashiCorp Vault. In-transit encryption uses TLS 1.3 with AES-GCM cipher suites.

Signed-off by Security Governance Board on January 22, 2026.